INCH-POUND
MIL-PRF-62461D(AT)
11 April 1997
SUPERSEDING
MIL-T-62461C(AT)
14 April 1989

PERFORMANCE SPECIFICATION

TRANSMISSION ASSEMBLY

This specification is approved for use by U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a transmission with an integral torque converter that consists of the main assembly and a control valve. The transmission assembly will be referred to, herein, by the term 'transmission'. The transmission is a counter-shaft type with external clutches and a six-speed self-contained unit with the necessary gearing, shafting, spacers, bearings and all other parts necessary for satisfactory operation of the vehicle on which it is used (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2520

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-50271	- Plate, Identification.

DEPARTMENT OF DEFENSE

MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Combat/Tactical Service.
MIL-C-22750	- Coating, Epoxy, High Solids.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-53039	- Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.
MIL-PRF-62462	- Engine, Diesel, Liquid-Cooled, 8 Cylinder: 4 Cycle, V-Type, 220 KW (295 HP).
MIL-V-62468	- Vehicle, Armored Combat Earthmover, M9 ACE.

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

DEPARTMENT OF DEFENSE

12331877	- Transmission Assembly.
13214E2449	- Control Valve.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

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NUCLEAR REGULATORY COMMISSION (NRC)

Code of Federal Regulations (CFR) - Title 10, Parts 30 and 40.

(Copies of the Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).

(Application for copies should be addressed to American National Standard Institute, 11 West 42nd Street, New York, NY 10036.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for this type of item. The material shall be free of defects which adversely affect appearance, performance, or serviceability of the finished product (see 4.5.1).

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs (see 4.5.1).

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3.2.2 Prohibited material. Asbestos, cadmium, and radioactive materials shall not be used in this item. Radioactive material is defined by CFR, Title 10, Parts 30 and 40, and other radioactive material in which specific activity is greater than 0.002 microcuries per gram or the activity per item equals or exceeds 0.01 microcuries (see 4.5.1).

3.2.3 Rubber components. Rubber components shall be ozone resistant to the degree specified in the applicable specification or drawing (see 4.5.1).

3.3 Design and construction. The design and construction of the transmission shall be as specified herein, and in accordance with Drawing 12331877. The construction of the control valve shall be in accordance with Drawing 13214E2449. The transmission shall have a single stage three element integral torque converter, shall be designed to accept input power from a diesel engine in accordance with MIL-PRF-62462 and shall transmit power and torque in accordance with MIL-V-62468 (see 4.5.2).

3.3.1 Accessories and equipment. All accessories and equipment shall be installed and properly adjusted. Electrical accessories and equipment including wiring and electrical connections shall be waterproof and shall conform to the applicable drawings (see 4.5.2).

3.3.2 Seals. Seals shall restrict entrance of all foreign material into bearings exposed to contamination during transmission operation under all conditions. Seals shall prevent lubricant leakage and water contamination of lubricants (see 4.5.2).

3.3.3 Oil. The transmission shall be filled to operating level with oil conforming to Grade 15W-40 of MIL-L-2104 when operating in ambient air temperature above 0 degrees Fahrenheit (°F) (minus (-) 18 degrees Celsius (°C)). Oil conforming to MIL-L-46167 shall be used when operating in ambient air temperature below 0°F (-18°C) (see 4.5.2).

3.3.4 Weight. The dry weight of the transmission shall be not more than 1350 pounds (lb) (612 kilograms (kg)) and shall be not more than 1400 lb (635 kg) with residual oil (see 4.5.2).

3.4 Performance.

3.4.1 Overall performance. The transmission shall perform as specified herein and on Drawing 12331877, under all conditions and speed ranges, when coupled to a 220 kilowatts (kW) at 2600 revolutions per minute (rpm) power source in accordance with MIL-PRF-62462.

3.4.1.1 Output speed. Minimum output speeds shall be in accordance with table I (see 4.5.3.1).

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TABLE I. Minimum output speeds.

No load input speed (±20 rpm)	Gear range	Forward speed (rpm minimum)	Reverse speed (rpm minimum)
1800	1st	320	344
1800	2nd	485	518
1800	3rd	776	
1800	4th	1170	
1800	5th	1999	
1800	6th	3003	

3.4.2 Oil pressure and flow. The transmission, with an input speed of 1800 ± 20 rpm, no load at the output, and a converter out temperature of 200 ± 20°F (93 ± 7°C), shall attain the oil pressures and flows indicated in table II (see 4.5.3.2).

TABLE II. Oil pressure and flows.

Direction and range	Input speed, rpm	Clutch pressure, psi (kPa)	Converter-out flow, gal/min (L/min)	Converter-out temp, °F (°C)	Lube temp, °F (°C)	Lube pressure, psi (kPa)
R-1,2	1800 ± 20	180-220 (1240-1517)	17.4 (68)	180-220 (82-104)	180-220 (82-104)	15-40 (103-276)
N	1800 ± 20	180-220 (1240-1517)	20.0 minimum (78)	180-220 (82-104)	180-220 (82-104)	15-40 (103-276)
F-1,2, 3,4,5, 6	1800 ± 20	180-220 (1240-1517)	17.4 minimum (68)	180-220 (82-104)	180-220 (82-104)	15-40 (103-276)

NOTE: psi = pounds per square inch; gal/min = gallons per minute
kPa = kilopascals; L/min = liters per minute
temp = temperature

3.4.2.1 Oil temperature. The transmission shall be capable of operating with converter output temperatures of up to 250°F (121°C) on a continuous basis and up to 280°F (138°C) on an intermittent basis. Testing of this requirement is limited to situations where first article testing is required. Normal acceptance testing shall be accomplished with the converter out temperature in the range of 180-220°F (82-104°C) (see 4.5.3.2.1).

3.4.2.2 Oil leakage. During an evaluation sequence or at completion of all evaluations, the transmission shall show no evidence of leakage in excess of a seep (see 4.5.3.2.2 and 6.3).

3.4.3 Environmental.

3.4.3.1 High temperature. The transmission shall operate through all gear ranges at an ambient temperature of 120°F (49°C) without damage or malfunction (see 4.5.3.3.1).

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3.4.3.2 Low temperature. The transmission shall operate through all gear ranges without benefit of solar radiation after being cold soaked to an ambient temperature of -26°F (-32°C) without damage or malfunction (see 4.5.3.3.2).

3.4.3.3 Leak decay. All transmission components, connections, gaskets, seals, and joints shall be free of leaks when completely submerged in water (see 4.5.3.3.3).

3.4.3.4 Steam and waterjet cleaning. The transmission shall withstand steam cleaning and high pressure waterjet washing without deterioration of seals, water leakage past seals or gaskets, or other defects. Paint removal shall not be considered a defect (see 4.5.3.3.4).

3.4.4 Durability. The transmission shall have a 50 percent (%) probability of completing the first 600 hours (h) of operation without replacement, rebuild or overhaul (see 4.5.1).

3.5 Finish. Unless otherwise specified (see 6.2), the transmission shall be cleaned, treated, primed and painted in accordance with good commercial practice. All parts of the transmission that are exposed to chemical contaminations shall be painted with a coating conforming to MIL-C-22750, MIL-C-46168, or MIL-C-53039 (see 4.5.2).

3.6 Identification marking. The transmission and each major component shall be marked as specified on the applicable drawings and shall include as a minimum the following (see 4.5.2):

a. National stock number (NSN).
b. Manufacturer’s serial number.
c. Manufacturer’s name and CAGE code.
d. U.S.

3.6.1 Nameplates. Unless otherwise specified (see 6.2), the nameplate and data plates shall conform to A-A-50271 (see 4.5.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspections (see 4.4).

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4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 73 ± 18°F (23 ± 8°C).
b. Barometric pressure: 28.5 + 2.0 inches Mercury (in. Hg) -3.0 in. Hg (725 + 50 millimeters (mm) Hg -75 mm Hg).
c. Relative humidity: 50 ± 30%.

4.3 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on preproduction or initial production samples as specified (see 3.1).

4.3.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of two transmissions. Preproduction inspection shall consist of the tests of table III and the examinations of table IV.

4.3.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select two from the first ten transmissions produced under the production contract for initial production inspection. Initial production inspection shall consist of the tests of table III and the examinations of table IV.

TABLE III. Classification of inspection.

Title	Requirement	Inspection	First article	CI (100%)
Output speed	3.4.1.1	4.5.3.1	X	X
Oil pressure & flow	3.4.2	4.5.3.2	X	X
Oil temperature	3.4.2.1	4.5.3.2.1	X	X
Oil leakage	3.4.2.2	4.5.3.2.2	X	X
High temperature	3.4.3.1	4.5.3.3.1	X	
Low temperature	3.4.3.2	4.5.3.3.2	X	
Leak decay	3.4.3.3	4.5.3.3.3	X	X
Steam & waterjet cleaning	3.4.3.4	4.5.3.3.4	X	

4.4 Conformance inspections (CI). CI shall include the tests of table III and the examinations of table IV.

4.4.1 Examination.

4.4.1.1 Sampling. Unless otherwise specified (see 6.2), transmissions shall be examined for the defects in table IV. Samples from an inspection lot shall be selected in accordance with ANSI/ASQC Z1.4.

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4.4.2 Tests (100%). Each transmission shall be subjected to the conformance tests specified in table III.

TABLE IV. Classification of defects.

Category	Defect	Method of examination
Major:		
101	Dimensions affecting interchangeability, out of tolerance (see 3.3 and 3.3.1).	SIE 1/
Minor:		
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3 and 3.3.1).	SIE
202	Nonconformance in design & construction (see 3.3 through 3.3.3).	Visual & SIE
203	Weight not as specified (see 3.3.4).	SIE
204	Improper finish, color or application (see 3.5).	Visual
205	Improper marking & nameplates (see 3.6 and 3.6.1).	Visual

1/ SIE = Standard Inspection Equipment.

4.5 Methods of inspection.

4.5.1 Materials and construction. Conformance to 3.2 through 3.2.3 and 3.4.4 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.5.2 Defects. Conformance to 3.3 through 3.3.4 and 3.5 through 3.6.1 shall be determined by examination for the defects listed in table IV. Examination shall be visual, tactile, or by measurement with SIE.

4.5.3 Performance tests.

4.5.3.1 Output speed. To determine conformance to 3.4.1.1, the transmission shall be tested on a test stand operated under the input speed and gear ranges specified in table I, and output speed determined.

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4.5.3.2 Oil pressure and flow. To determine conformance to 3.4.2, the transmission shall be tested on a test stand with provisions for providing the no-load speed, gear shifting, and oil supply at the temperature indicated in table II. The test stand shall provide for measuring flows and pressures indicated.

4.5.3.2.1 Oil temperature. To determine conformance to 3.4.2.1, during each transmission evaluation sequence and production run, the transmission shall show capability of operating with oil at the temperatures indicated without damage or leakage beyond a seep for a period of not more than 15 minutes (see 6.3).

4.5.3.2.2 Oil leakage. To determine conformance to 3.4.2.2, the transmission shall operate under the conditions specified in 4.5.3.2 and checked for any evidence of leakage in excess of a seep (see 6.3).

4.5.3.3 Environmental.

4.5.3.3.1 High temperature. To determine conformance to 3.4.3.1, the transmission, without external load and with provisions for oil cooling, shall be tested as follows:

a. Place the transmission in a test chamber for not less than 6 h at a temperature of 120°F (49°C).
b. Raise the chamber temperature to 160°F (71°C) within a time period of 1 h and maintain at that temperature for an additional 4 h.
c. Lower the chamber temperature to 120°F (49°C) within 1 h.
d. Repeat steps a, b, and c two additional times (making a total of three 12 h cycles).
e. Adjust the chamber temperature to the highest operating temperature under which the transmission is designed to operate and maintain until temperature stabilization of the transmission is reached.
f. Operate the transmission until it is stabilized through all gear ranges.
g. Return the transmission, nonoperating, to standard ambient conditions and stabilize.

4.5.3.3.2 Low temperature. To determine conformance to 3.4.3.2, the transmission, without external load and with provisions for oil cooling, shall be tested as follows:

a. Place the transmission in a test chamber for not less than 24 h at a temperature of -26°F (-32°C).
b. Raise the chamber temperature to -20°F (-30°C) and maintain until temperature stabilization of the transmission is reached.
c. Operate the transmission until it is stabilized through all gear ranges.

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d. Return the transmission, nonoperating, to standard ambient conditions and stabilize.
e. The rate of temperature change (steps a, b, and d) may be the maximum attainable by the chamber but shall not be greater than 18°F (10°C) per minute.

4.5.3.3.3 Leak decay. To determine conformance to 3.4.3.3, the transmission, without oil, shall be submerged in water to a depth sufficient to completely cover the assembly for not less than 5 minutes. The internal air pressure shall be 3-5 psi (20-35 kPa). The transmission shall be checked for leaks. No air bubbles are permissible.

4.5.3.3.4 Steam and waterjet cleaning. To determine conformance to 3.4.3.4, all transmission interface openings shall be sealed and the transmission shall be steam cleaned using high pressure steam and cleaner. After steam cleaning the transmission shall be waterjet washed and subsequently checked for any deterioration of seals or water leakage past seals or gaskets. Following the steam and waterjet cleaning, the oil shall be drained from the transmission and examined for contamination and may be refilled into the transmission if not contaminated.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The transmission covered by this specification is intended for use as original, spare, or replacement in the Armored Combat Earthmover M9 ACE.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Drawing number of the transmission.
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).

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d. If first article inspection is required (see 3.1).
e. If cleaning, treating, priming and painting of the transmission should be other than as specified (see 3.5).
f. If nameplates and data plates are other than as specified (see 3.6.1).
g. If inspection conditions are other than as specified (see 4.2).
h. If preproduction inspection is required (see 4.3.1).
i. If preproduction sample size should be other than as specified (see 4.3.1).
j. If initial production inspection is not required (see 4.3.2).
k. If initial production sample size should be other than as specified (see 4.3.2).
l. If sampling plan is other than as specified (see 4.4.1.1).
m. Packaging requirements (see 5.1).

6.3 Leakage. The following definitions are established as guidelines governing leakages.

a. Weep. Any evidence of fluid beyond a seal or joint.
b. Seep. Any evidence of fluid beyond a seal or joint that does not result in the formation of a droplet.
c. Leak. Any evidence of fluid beyond a seal or joint that results in the formation of a droplet.
d. Drip. Any evidence of fluid beyond a seal or joint where droplets form and fall.

6.4 Subject term (key word) listing.

Armored combat earthmover
M9 ACE
Power train

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - AT

Preparing Activity:
Army - AT

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-62461D(AT)	**2. DOCUMENT DATE *(YYMMDD)*** 970411

3. DOCUMENT TITLE
Transmission Assembly

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	**b. ORGANIZATION**	
c. ADDRESS *(Include Zip Code)*	**d. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** ***(If applicable)***	**7. DATE SUBMITTED *(YYMMDD)***

8. PREPARING ACTIVITY

a. NAME	**b. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** **(810) 574-8745** **786-8745**
c. ADDRESS *(Include Zip Code)* **Commander** **U.S. Army Tank-automotive and Armaments Command** **ATTN: AMSTA-TR-E/BLUE** **Warren, MI 48397-5000**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403** **Falls Church, VA 22041-3466** **Telephone (703) 756-2340 AUTOVON 289-2340**

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